

Brain to Text

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Abstract

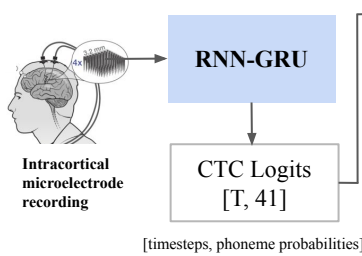
Building upon the 2024 research study “**An Accurate and Rapidly Calibrating Speech Neuroprosthesis**,” this project focuses on optimizing a neural decoding model that translates intracranial EEG (iEEG) signals into intended sentences. Paired with BCIs, such models enable individuals with speech impairments to communicate more effectively. The baseline model architecture provided from the study uses an RNN-GRU to compute phoneme probabilities, which are then fed into an N-Gram and LLM rescoring pipeline to generate final sentence predictions. Our optimization approaches incorporate methods such as Temporal Masking, the usage of Diphones instead of phonemes, Rebuilding the CTC decoding pipeline and using Cold Fusion LLM. Given sufficient computational resources and compatible OS environments, future work could explore strategies including larger N-Gram models, Model Ensembling and conducting more comprehensive training, validation, and testing experiments across model variations.

Literature Review & Introduction

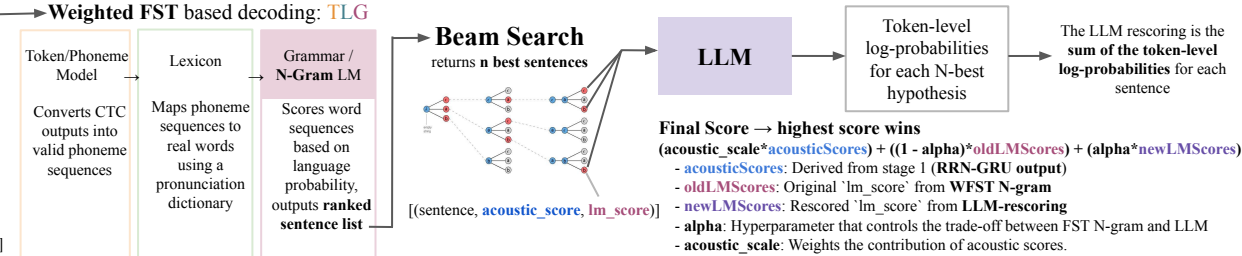
- Neurological conditions such as ALS can impair a person’s ability to speak. Modern BCIs are capable of decoding intended speech directly from neural activity, which can then be converted into text or synthesized speech, enabling communication. [1], [2].
- The first high-performance word-decoding speech BCI was demonstrated by Moses et al. [2]. They achieved a median word error rate (WER) of 25.6% at a speech rate of 15.2 words per minute (WPM).
- Willett et al. [1] later improved upon this by leveraging Utah arrays and a more advanced decoding framework, reaching a mean WER of 24% at 75 WPM.
- The Brain-to-Text Benchmark project [4] further expanded this task by incorporating LLMs, achieving a mean WER of 2.5% at a speech rate of 31.6 WPM. This work serves as the foundation of our project.
- Our dataset consists of recordings from a 45-year-old male ALS patient with tetraparesis and severe dysarthria, who was implanted with four microelectrode arrays in the left ventral precentral gyrus five years after symptom onset [2].
- The baseline model achieves a WER of 5.77% [4], and **our goal is to modify and fine-tune the pipeline to further reduce this error.**

Baseline Model Architecture

Stage 1: RNN-GRU



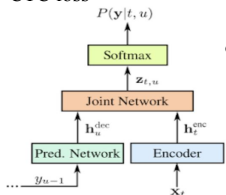
Stage 2: N-Gram + LLM Rescoring



Optimizations & Replacements

Stage 1: RNN-GRU

- Added **Temporal Masking** (Frequency and Random)
 - intentionally hide parts of the spectrogram during training so the model learns to be robust to noise, missing data, and artifacts
- Frequency Masking** → hide random horizontal bands
- Random Masking** → hide random vertical bands
- Used **RNNT + RNNT loss** instead of GRU and instead of CTC loss

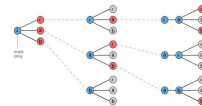


- diphone training** instead of phonemes
 - diphone** = pair of phonemes, transitions between sounds

Stage 2: N-Gram + LLM Rescoring

- Rebuilt CTC decoding pipeline** → beam search + lexicon + beam search

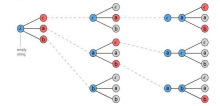
Ran a phoneme-level beam search over the CTC logits to generate high-probability phoneme sequences.



Applied a phoneme → word lexicon to prune invalid or impossible word mappings

/K AE T/ → CAT (valid)
/K AE P/ → CAP (valid)
/K AE TH/ → (no matching word → discarded)

Ran a second word-level beam search to generate refined, linguistically valid sentence hypotheses



- Used a **diphone-based lexicon**
 - mapped diphones → words instead of mapping single phonemes → words
- Experimented with **different sized N** → number of sentences passed through N-Gram and scored by LLM
- Using better LLM for rescoring → **LLaMA 3.3**
- Used **Cold fusion LLM** implementation
 - Idea: Freeze LLaMA 3.3 → train a GRU/Transformer decoder that combines phoneme CTC embeddings with LLaMA’s hidden state

Results and Future Directions

More compute, implementing the decoder in C++, or switching fully to Linux, would allow:

- Using **larger N-Gram models** (we are currently limited to bigram)
- Faster RNN training**, especially for diphone models
- Building a **WFST graph for the lexicon + N-gram** (baseline used this, but our OS setup prevented it)
- Running **more extensive training, validation, and test experiments** across model variations
- Performing dynamic **hyperparameter tuning**
- Model Ensembling**: train 41 smaller RNNs (one per phoneme), a previously winning approach
- Implementing a **Conformer architecture** as outlined by Song et al. (2023).

References

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